

# Full Deployment Guide

## EC2 Web Server with RDS on AWS (Terraform)



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## Project Overview

This code creates an integrated infrastructure on AWS using Terraform to automatically manage and configure cloud computing resources.

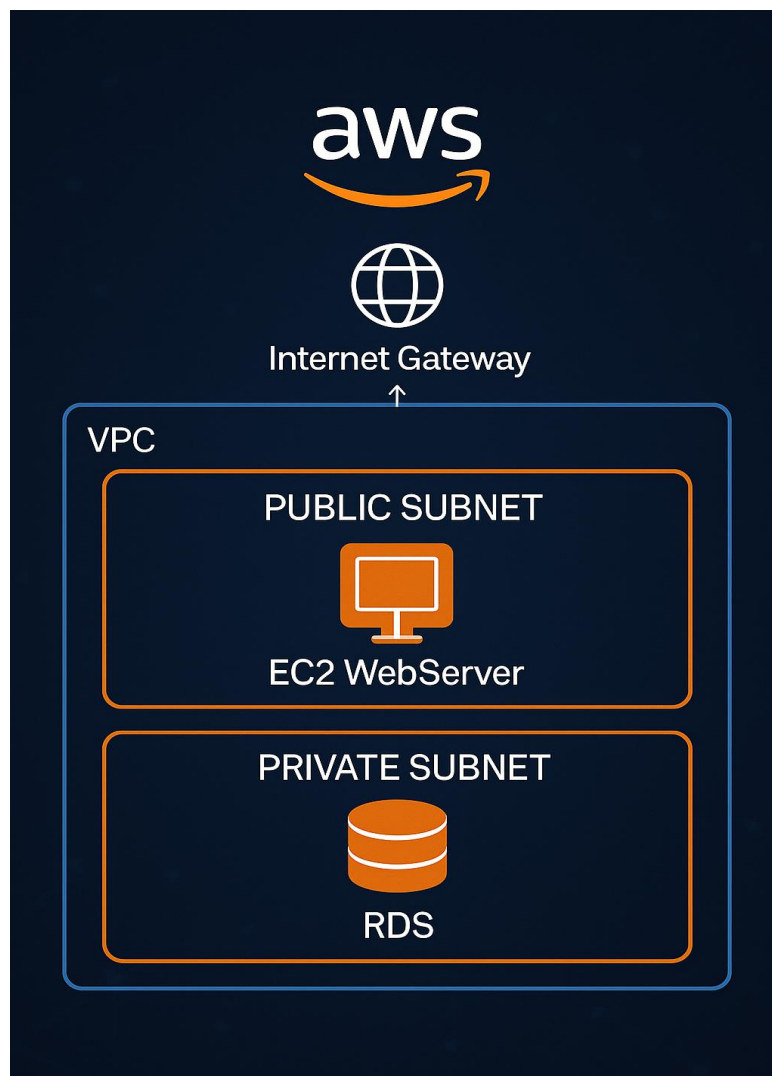
The code includes setting up a virtual private network (VPC) with public and private subnets, creating internet gateways and NAT to securely connect the network to the internet.

The code also creates a MySQL database using RDS, and an EC2 server configured to run a Node.js web application and connect directly to the database.

In addition, an RSA key is generated and an AWS key pair is configured for secure login on the server. S3 is used to store Terraform state files to ensure secure and organized change management.

This code provides a flexible, scalable, and manageable infrastructure without the need for manual intervention, helping accelerate application deployment and deployment.

## Architecture Diagram:



## Step 1: AWS provider ( provider.tf )

Terraform uses the AWS provider (hashicorp/aws) **version 5.36.0**. The AWS provider defines the access keys (**access\_key** and **secret\_key**) and the **us-east-1** region.

## Step 2: networking ( vpc.tf )

### 1-VPC and Subnets

Creates a VPC named **`myvpc`** with the address **`10.0.0.0/16`**, containing a **public subnet** (**`10.0.0.0/24`**) and a **private subnet** (**`10.0.1.0/24`**) in two different zones.

Your VPCs (3) Info

Last updated less than a minute ago

Actions

Create VPC

Find VPCs by attribute or tag

|                          | Name  | VPC ID                                | State     | Block Public... | IPv4 CIDR   | IPv6 CIDR |
|--------------------------|-------|---------------------------------------|-----------|-----------------|-------------|-----------|
| <input type="checkbox"/> | myvpc | <a href="#">vpc-000a534ea73392b93</a> | Available | Off             | 10.0.0.0/16 | -         |

Subnets (5) Info

Last updated 1 minute ago

Actions

Create subnet

Find subnets by attribute or tag

|                          | Name           | Subnet ID                                | State     | VPC                                           | Block Public... | IPv4 CIDR   |
|--------------------------|----------------|------------------------------------------|-----------|-----------------------------------------------|-----------------|-------------|
| <input type="checkbox"/> | public-subnet  | <a href="#">subnet-035beddaa116dd898</a> | Available | <a href="#">vpc-000a534ea73392b93</a>   myvpc | Off             | 10.0.0.0/24 |
| <input type="checkbox"/> | private-subnet | <a href="#">subnet-0e5075a1efa5e989d</a> | Available | <a href="#">vpc-000a534ea73392b93</a>   myvpc | Off             | 10.0.1.0/24 |

### 2-Internet Gateway

Adds an internet gateway connected to the VPC to provide internet connectivity to the public subnet.

Internet gateways (2) Info

Last updated 4 minutes ago

Actions

Create internet gateway

Find internet gateways by attribute or tag

|                          | Name | Internet gateway ID                   | State    | VPC ID                                        | Owner        |
|--------------------------|------|---------------------------------------|----------|-----------------------------------------------|--------------|
| <input type="checkbox"/> | igw  | <a href="#">igw-09b360d349b9e8bd0</a> | Attached | <a href="#">vpc-000a534ea73392b93</a>   myvpc | 337909736881 |

### 3-Route Tables

Sets up a public route table (connects to the internet via an IGW) and a private route table (connects to the internet via a NAT gateway).

Route tables (5) Info

Last updated 4 minutes ago

Actions

Create route table

Find route tables by attribute or tag

|                          | Name                | Route table ID                        | Explicit subnet associ...                | Edge associations | Main | VPC                                   |
|--------------------------|---------------------|---------------------------------------|------------------------------------------|-------------------|------|---------------------------------------|
| <input type="checkbox"/> | -                   | <a href="#">rtb-0aeac153722d29aee</a> | -                                        | -                 | Yes  | <a href="#">vpc-0d326bb98a60ca447</a> |
| <input type="checkbox"/> | private-route-table | <a href="#">rtb-0ad473d886cee637e</a> | <a href="#">subnet-0e5075a1efa5e9...</a> | -                 | No   | <a href="#">vpc-000a534ea73392b93</a> |
| <input type="checkbox"/> | public-route-table  | <a href="#">rtb-0d36bac299f252516</a> | <a href="#">subnet-035beddaa116dd...</a> | -                 | No   | <a href="#">vpc-000a534ea73392b93</a> |

## 4-Route Associations

Associates each subnet with the correct route table: the public subnet to the public route table, and the private subnet to the private route table.

## 5-Elastic IP and NAT Gateway

Elastic allocates an IP and builds a NAT Gateway within the public subnet to allow the private subnet to exit to the internet.

NAT gateways (2) Info

Find NAT gateways by attribute or tag

Actions

Create NAT gateway

< 1 >

|                       | Name | NAT gateway ID                        | Connectivity... | State                  | State message | Primary public I...           | Primary priva |
|-----------------------|------|---------------------------------------|-----------------|------------------------|---------------|-------------------------------|---------------|
| <input type="radio"/> | -    | <a href="#">nat-062b95f3816b7fb77</a> | Public          | <span>Available</span> | -             | <a href="#">13.219.38.129</a> | 10.0.0.48     |

## Step 3: Security Groups ( security\_group.tf )

- 1-Creates a Security Group for the RDS connection named ``rds-sg``, and one for EC2 named ``web-sg`` in the same VPC.
- 2-Temporarily opens all inbound traffic (protocols + ports) to the web server using ``-1``.
- 3-Opens all outbound traffic from the web server to any location.
- 4- Adds a special rule in RDS that opens port **3306** (MySQL) only for servers in ``web-sg``.
- 5-Adds an explicit rule that opens port **3000** for the Node.js app to receive requests from the internet.
- Only allows EC2 (not public) to access the database.

|                          |   |                                      |         |                                       |                   |
|--------------------------|---|--------------------------------------|---------|---------------------------------------|-------------------|
| <input type="checkbox"/> | - | <a href="#">sg-04fc122a4909f1e02</a> | rds sg  | <a href="#">vpc-000a534ea73392b93</a> | SG for RDS        |
| <input type="checkbox"/> | - | <a href="#">sg-074c79b9bd24143c3</a> | default | <a href="#">vpc-0e046182b24321f1f</a> | default VPC secur |
| <input type="checkbox"/> | - | <a href="#">sg-0354afd9ce4abd785</a> | web sg  | <a href="#">vpc-000a534ea73392b93</a> | SG for Web Serve  |

## Step 4: Launch EC2 Instance ( ec2.tf )

- A web server inside the public subnet.
- Replace the AMI with your region-specific **Amazon Linux 2 AMI**.
- Generates a 4096-bit **RSA key** using ``tls_private_key`` and saves a local copy of the private key in the ``terraform-key.pem`` file.
- Creates an **AWS key pair** using the generated public key and binds it to EC2 to allow SSH connections.
- Creates an EC2 instance on a **public subnet**, bound to the appropriate security group (``web-sg``), and with a public IP address for external access.
- Provisioning Within ``remote-exec``, it sets environment variables, downloads all required software, modifies ``index.js`` to listen to ``0.0.0.0``, and runs the application in the background.

Instances (1)

Info

Last updated less than a minute ago

Connect

Instance state ▾

Actions ▾

Launch instances ▾

Find Instance by attribute or tag (case-sensitive)

Running ▾

< 1 >

| <div><input type="checkbox"/></div> <div>Name <div></div></div> | Instance ID         | Instance state <div></div>                                       | Instance type <div></div> | Status check                             | Alarm status             | Availability Zone <div></div> |
|-----------------------------------------------------------------|---------------------|------------------------------------------------------------------|---------------------------|------------------------------------------|--------------------------|-------------------------------|
| <div><input type="checkbox"/></div>                             | i-080a106a9f7620d9a | <div><div></div></div> Running <div><div></div><div></div></div> | t2.micro                  | <div><div></div></div> 2/2 checks passed | <div>View alarms +</div> | us-east-1a                    |

## Step 5: Setup RDS MySQL in Private Subnet ( rds.tf )

1-The code creates an RDS **MySQL** Database named ``islammyrds`` using **MySQL 5.7** and stores its data in a **20 GB** size with the ``gp2`` storage type.

2-t uses a small instance type ``db.t3.micro`` and is linked to the security group ``rds-sg`` to control access.

3-It specifies the RDS running location at ``us-east-1b``

4-It relies on a subnet group called ``myrds-subnet-group``, which includes the public and private subnets to distribute the network.

5-It determines the username and password for the database from Terraform variables (``var.MYSQL_USER`` and ``var.MYSQL_PASSWORD``) and uses ``default.mysql5.7`` as the parameter group.

islammyrds

Modify

Actions

Summary

DB identifier

islammyrds

CPU

3.52%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

1 Connections

Engine

MySQL Community

Region & AZ

us-east-1b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Recommendations

Connectivity & security

Endpoint & port

Endpoint

islammyrds.cyh6c442myne.us-east-1.rds.amazonaws.com

Port

3306

Networking

Availability Zone

us-east-1b

VPC

myvpc (vpc-000a534ea73392b93)

Subnet group

myrds-subnet-group

Subnets

subnet-0e5075a1efa5e989d  
subnet-035beddaa116dd898

Network type

Security

VPC security groups

rds.sg (sg-04fc122a4909f1e02)

Active

Publicly accessible

No

Certificate authority

Info

rds-ca-rsa2048-g1

Certificate authority date

May 26, 2061, 02:34 (UTC+03:00)

DB instance certificate expiration date

April 30, 2028, 23:16 (UTC+03:00)

## Step 6: S3 ( s3.tf )

The code creates an S3 bucket named `s3-bucket-t000-backend`` to be used as a backend for storing state. It adds tags to define the name (``My bucket``) and environment (``Dev``).

### s3-bucket-t00-backend [Info](#)

ObjectsMetadataPropertiesPermissionsMetricsManagementAccess Points

Objects (1)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

< 1 >

| <input type="checkbox"/> | Name                                                                                                                  | Type    | Last modified                     | Size    | Storage class |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|---------|-----------------------------------|---------|---------------|
| <input type="checkbox"/> |  <a href="#">terraform.tfstate</a> | tfstate | May 1, 2025, 23:12:00 (UTC+03:00) | 181.0 B | Standard      |

## Step 7: Backend ( backend.tf )

A Terraform backend stores state files in an S3 bucket named `s3-bucket-t000-backend`` with a key of `terraform.tfstate``. This makes them available to the entire team for collaborative work.

# Final result

Not secure 3.208.73.203:3000



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😊 Thank You 😊